Assignments

Topic-wise practice worksheets, one folder per week of the plan in `../docs/03-practice-plan.md`. Each folder (once built) contains:

- `assignment.md` — the questions (4-option MCQ, Class 4 difficulty, mixing direct computation with word-problem/Everyday-Maths style framing)
- `answer-key.md` — correct options with a short worked explanation each

Folder list (per the 12-week plan)

Folder	Topic	Status
`01-number-sense/`	Place value, number names, ordering, rounding, Roman numerals	✅ Built
`02-four-operations/`	Addition, subtraction, multiplication, division (multi-digit + word problems)	✅ Built
`03-factors-multiples-patterns/`	Factors, multiples, number patterns	✅ Built
`04-fractions-decimals/`	Comparing & equivalent fractions, simple decimals	✅ Built
`05-measurement-money/`	Length, weight, capacity, money, conversions	✅ Built
`06-time-calendar/`	Clock time, duration, calendar	✅ Built
`07-geometry-perimeter-area/`	Shapes, symmetry, perimeter, basic area	✅ Built
`08-data-handling/`	Tally marks, pictographs, bar graphs	✅ Built
`09-reasoning-patterns-coding/`	Number/figure patterns, analogy, classification, alphabet test, coding–decoding	✅ Built
`10-reasoning-direction-figures/`	Ranking, direction sense, days & dates, mirror images, embedded figures, geometrical solids	✅ Built (24 Q)
`11-achievers-hots/`	Multi-step HOTS problems across all topics above	✅ Built
`12-math-kangaroo-stretch/`	**Optional** — Math Kangaroo-style puzzles (shape counting, mirror images, logic, cube nets)	✅ Built (15 Q, 3 tiers)
`13-achievers-level2/`	**Level-2 track** — harder, multi-step HOTS beyond Week 11, for students who've qualified Level 1	✅ Built

Folder	Topic	Status
`14-reasoning-level2/`	**Level-2 track** — compound logical-reasoning puzzles stacking 2+ clues/rules, beyond Weeks 9–10	✅ Built
`15-geometry-level2/`	**Level-2 track** — compound-shape perimeter & area (paths, frames, cut corners), beyond Week 7	✅ Built
`16-indian-international-numbers/`	Indian vs. International place-value systems, period names, comma placement, cross-system conversion — identified as a gap against the school's own syllabus, not yet slotted into the 12-week schedule	✅ Built
`17-indian-international-numbers-achievers/`	**Achievers/HOTS** — multi-step conversion + arithmetic, beyond Folder 16	✅ Built
`18-indian-international-numbers-level2/`	**Level-2 track** — crores/hundred-millions, multi-step conversions, beyond Folder 16	✅ Built
`19-subtracting-figures/`	**Figure Subtraction** reasoning puzzles (Figure 1 − Figure 2 = Figure 3) — identified as a gap against the school's own syllabus, not yet slotted into the 12-week schedule	✅ Built (15 Q, all diagram-based)
`20-subtracting-figures-achievers/`	**Achievers/HOTS** — two-part subtractions plus reversed "find Figure (1)" puzzles, beyond Folder 19	✅ Built (15 Q, all diagram-based)
`21-inserting-missing-numbers/`	**Inserting Missing Numbers** reasoning puzzles — number grids and pyramids with a hidden rule — identified as a gap against the school's own syllabus, not yet slotted into the 12-week schedule	✅ Built (15 Q)
`22-inserting-missing-numbers-achievers/`	**Achievers/HOTS** — column-direction patterns, magic-square grids, two-step rules, number pyramids, beyond Folder 21	✅ Built (15 Q)
`23-incomplete-pattern/`	**Incomplete Pattern** reasoning puzzles — 3×3 figure grids with a hidden rule, find the missing figure — identified as a gap against the school's own syllabus, not yet slotted into the 12-week schedule	✅ Built (15 Q, all diagram-based)
`24-incomplete-pattern-achievers/`	**Achievers/HOTS** — two-axis (row + column) rules and true Latin-square grids, including questions where the missing cell isn't in the bottom-right corner, beyond Folder 23	✅ Built (15 Q, all diagram-based)

Folder	Topic	Status
`25-fraction-interpretation/`	**Fraction Interpretation** — shaded-area fractions, fraction of a group of discrete objects, fraction of a quantity, mixed-number ↔ improper-fraction conversion — identified as a depth gap beyond Week 4 (`04-fractions-decimals/`), not yet slotted into the 12-week schedule	✅ Built (15 Q)
`26-fraction-interpretation-achievers/`	**Achievers/HOTS** — multi-step fraction-of-quantity word problems, unlike-denominator comparison/addition, mixed-number arithmetic, and visual equivalence/misconception-trap diagrams, beyond Folder 25	✅ Built (15 Q)

Folders `01` – `11` are the core 12-week plan, built in the same two-file (`assignment.md` + `answer-key.md`) shape, 15 questions each — except `10-reasoning-direction-figures/` , which has 24 (15 text-based + 9 diagram-based). `12-math-kangaroo-stretch/` is the optional extension described in `docs/03-practice-plan.md` , not part of the core plan or any Indian olympiad's syllabus. `08-data-handling/` , `10-reasoning-direction-figures/` , and `12-math-kangaroo-stretch/` each ship an `images/` folder of SVG diagrams referenced from their `assignment.md` .

Folders `13` – `15` are a **Level-2 topic track**: for students who've qualified Level 1 of the real exam and want harder, topic-focused practice rather than just the mixed-topic Level-2 sample papers (`sample-papers/sample-paper-06.md` onward). Same two-file shape, 15 questions each, every answer verified programmatically. These three topics were picked because they're where Level 2 gets *conceptually* harder, not just numerically bigger — the other core topics (number sense, four operations, data handling, etc.) are already covered by the harder numbers in the Level-2 sample papers.

Diagram-based question types

Mirror images, embedded figures, and geometrical-solid recognition are inherently visual Section-1 (Logical Reasoning) sub-topics. Rather than describing them in text, `10-reasoning-direction-figures/images/` has an SVG diagram for each question:

- **Mirror images** — the original figure plus 4 candidate reflections (one correctly flipped left-right, the other three are unchanged / flipped upside-down / rotated 180° — classic distractors). The correct option is generated by an exact `scale(-1,1)` transform, not hand-drawn, so it's guaranteed accurate.
- **Embedded figures** — a composite figure made of two overlapping shapes; find which of 4 candidate shapes is traceable inside it.

- **Geometrical solids** — a drawn cube, cone, and cylinder, each with a question about its name or face count.

`08-data-handling/images/` similarly has a real SVG pictograph and bar graph (rather than an emoji/markdown-table stand-in) for its data-handling questions.

`12-math-kangaroo-stretch/images/` has 7 more SVGs for its puzzle questions — shape-counting figures, a mirror-image puzzle (same guaranteed-correct transform technique as above), a balance-scale illustration, a grid for path-counting, a shaded-fraction figure, and a labelled cube net.

`19-subtracting-figures/images/` and `20-subtracting-figures-achievers/images/` have an SVG for every question (30 total) — **Figure Subtraction** puzzles, where each figure is built from named parts (an outline, diagonals, a flap, a circle...) and Figure (2) is always drawn from a literal sub-list of Figure (1)'s own parts, so the correct "Figure (1) − Figure (2)" remainder is a guaranteed-correct set difference, not a hand-judged drawing. Folder 20 adds two-part subtractions and a few reversed puzzles (given what was subtracted and what remained, work out the original figure).

`23-incomplete-pattern/images/` and `24-incomplete-pattern-achievers/images/` have an SVG for every question (30 total) — **Incomplete Pattern** 3×3 figure-matrix puzzles. Every figure is generated from pure geometry (dot/star/line counts, regular-polygon side counts via trig, square scale, rotation angle, shading fraction) rather than hand-drawn, so each grid's rule — and the one correct missing figure — is guaranteed-correct by construction and independently re-verified against every one of the 9 cells (not just the ones shown) before publishing. Folder 23 uses a single column-wise rule repeated down every row; Folder 24 adds genuine two-axis (row **and** column) rules and true Latin-square grids, including a few questions where the missing cell isn't in the bottom-right corner.

`25-fraction-interpretation/images/` and `26-fraction-interpretation-achievers/images/` have an SVG for every diagram-based question — rectangle/grid and pie-chart area models, groups of discrete shapes (including mixed shape-type groups for "fraction of a category" questions), and a few "which diagram is correct" questions with 4 candidate diagrams as the answer options, including the classic unequal-parts misconception trap. All partitions are computed geometrically (exact trig for pie sectors), never hand-drawn.
